

1 Type I calibration of fixed-effect interaction tests in linear
2 mixed models: a staged diagnosis of working-covariance
3 misspecification and a cluster-robust remedy

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Abstract

Background. The linear mixed model is the standard tool for testing a fixed-effect interaction in longitudinal and clustered data. The model-based Wald test of that interaction is correctly calibrated only when the working within-cluster covariance is correctly specified. In practice an analyst chooses a convenient parametric residual structure, a continuous-time autoregressive correlation, compound symmetry, or an unstructured form, and then trusts the model-based standard error. The direction and the magnitude of the miscalibration that follows when the working covariance is wrong are rarely checked.

Methods. We set out a staged diagnostic protocol, a four-channel decomposition of the test statistic, a working-covariance model ladder, and a sample-size gradient, and apply it to a worked example: the test of a biomarker-treatment interaction in an aggregated N-of-1 carryover simulation, analyzed with an `lme` model carrying a random intercept and a `corCAR1` residual correlation.

Results. At the null the model-based standard error of the interaction is over-estimated by 6 to 10 percent, and the realized Type I error is approximately 0.03 against a nominal 0.05. The degrees-of-freedom rule, the positive-definiteness correction, and point-estimate bias are excluded as causes. The model ladder localizes the miscalibration to the `corCAR1` residual layer: a random intercept alone is well calibrated. The sample-size gradient shows the bias is asymptotic, not a moderate-sample artifact: it does not attenuate from $N = 35$ to $N = 280$, and an empirical Kenward-Roger check confirms the small-sample correction is numerically inert on the parsimonious autoregressive covariance. A CR2 bias-reduced cluster-robust standard error restores the standard-error ratio to approximately one and the Type I error to approximately 0.05; a generalized-estimating-equations analysis with a bias-corrected sandwich reaches the same calibration from the marginal side. At an alternative cell with a non-null effect the recalibration raises power by a near-uniform increment and leaves the ranking of analysis specifications unchanged.

Conclusions. A misspecified parametric working covariance can bias the model-based standard error of a mixed-model interaction test asymptotically, in either direction. Cluster-robust standard errors are a general remedy, and the staged diagnostic is a reusable instrument for detecting the problem.

1 Introduction

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The linear mixed model is, by a wide margin, the most common vehicle for testing whether an effect is moderated by a covariate in longitudinal or clustered data^{1,2}. The quantity of interest is a fixed-effect interaction coefficient, and the inference almost always proceeds through a Wald test: the coefficient estimate is divided by a model-based standard error and referred to a t distribution. That procedure is correctly calibrated, exactly so in balanced cases and to first order otherwise,

only under a condition that is easy to state and, unfortunately, easy to forget. The working covariance, that is, the random-effects structure together with the parametric residual correlation the analyst has assumed, must be correct.

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It has been understood since the work of Eicker, Huber, and White that this condition is more demanding than it appears³⁻⁵. When the working covariance is misspecified, the model-based standard error is no longer a consistent estimator of the true sampling variability of the fixed-effect estimate. Liang and Zeger⁶ carried this insight into longitudinal data analysis: their generalized estimating equations decouple the point estimate, computed under a convenient working correlation, from the variance estimate, computed by an empirical sandwich that remains consistent whatever the working correlation happens to be. This cluster-robust (sandwich) standard error has since become a standard instrument, and a substantial literature has refined it for the small-sample case⁷⁻¹⁰.

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Yet in routine mixed-model practice the working covariance is still trusted. An analyst fitting longitudinal data with `nlme` or a comparable package will choose a residual correlation, very often a continuous-time first-order autoregressive form, because it is pharmacokinetically or biologically plausible¹¹, and will then report the model-based standard error of the interaction without asking what happens if the chosen form is not exactly the form the data follow. This is not negligence so much as a reasonable working assumption that is seldom audited. But how large is the error when the assumption fails? Is the resulting test conservative or anti-conservative? Does the miscalibration diminish as the sample grows, so that it can perhaps be dismissed as a small-sample nuisance, or does it persist? These questions do not have a single answer, because the answer depends on the interaction being tested and on the way the working covariance departs from the truth. They have to be asked of a particular analysis, and we shall ask them of one.

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In this paper we endeavor to do exactly that for one analysis, and in doing so we offer two contributions that we believe are more general than the single case. The first is a diagnostic protocol. We decompose the Wald statistic into four channels through which miscalibration can enter, and we add two further instruments: a working-covariance model ladder that localizes the cause, and a sample-size gradient that separates a finite-sample artifact from an asymptotic one. The protocol is not specific to the example and can be applied to any mixed-model interaction test whose calibration is in doubt. The second contribution is the finding

the protocol produced. In our worked example the model-based interaction test is conservative, not anti-conservative as the sandwich literature’s emphasis on underestimated standard errors might lead one to expect; the conservatism is material, on the order of ten percent in standard-error terms; and it is asymptotic, persisting undiminished across an eightfold range of sample size. We trace it to a single component of the working covariance, and we show that a cluster-robust standard error removes it.

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The worked example is drawn from the pmsimstats-ng project, a simulation study of power to detect biomarker-treatment interactions in aggregated N-of-1 trials under carryover¹². The example is incidental to the argument. The phenomenon, a parametric working covariance that biases a fixed-effect interaction test asymptotically, and the remedy are not properties of that simulation; they are properties of the mixed-model Wald test, and the example serves only to make them concrete and quantified. We begin in Section 2 with the diagnostic protocol. Section 3 describes the worked example. Section 4 reports the staged diagnosis and the cluster-robust remedy. Section 5 discusses the generality of the result and its limits.

2 A staged diagnostic protocol

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Consider a fixed-effect interaction coefficient θ estimated by $\hat{\theta}$, with a model-based standard error $\widehat{\text{SE}}$ and an assigned degrees of freedom ν . The Wald test rejects the null $H_0 : \theta = 0$ at level α when the statistic $T = \hat{\theta}/\widehat{\text{SE}}$ exceeds the critical value $t_{\nu, 1-\alpha/2}$ in absolute value. The test is correctly calibrated when the realized rejection rate under H_0 equals α . When it does not, the discrepancy must enter through one of exactly four channels, and the channels are separable by simulation. The general anatomy of a Wald statistic and of the reference distribution against which it is judged is itself classical¹³. We begin with the decomposition itself, and then describe the two further instruments we shall add to it.

2.1 The four-channel decomposition

We begin by writing the interaction estimate in a form that exposes the four channels. We use the word ‘channel’ for the four mutually exclusive routes by which a discrepancy can reach the realized error rate; the term is adopted because the more familiar nouns are already committed in this setting, ‘component’ to variance components, ‘factor’ to the latent factors of the data-generating process, and ‘term’ to the terms of the model. Let

$$\hat{\theta} = \mu + \sigma Z^*, \quad \mu = \text{E}(\hat{\theta}), \quad \sigma = \text{SD}(\hat{\theta}), \quad (1)$$

so that $Z^* = (\hat{\theta} - \mu)/\sigma$ is the standardized estimate, of mean zero and unit variance by construction; when $\hat{\theta}$ is approximately normal, Z^* is approximately standard normal. Write $\widehat{\text{SE}}$ for the mean of the model-based standard error and $\kappa = \widehat{\text{SE}}/\sigma$ for its ratio to the true sampling standard deviation σ . Then the Wald statistic, to the order at which the variability of $\widehat{\text{SE}}$ about its mean may be neglected, is

$$T = \frac{\hat{\theta}}{\widehat{\text{SE}}} \approx \frac{\mu + \sigma Z^*}{\kappa \sigma} = \frac{1}{\kappa} (Z^* + \delta), \quad \delta = \frac{\mu}{\sigma}, \quad (2)$$

where δ is the bias expressed in standard-deviation units. In this form the statistic is a shifted and rescaled standard normal, the structure that underlies the noncentral t distribution¹⁴: δ enters as a non-centrality and κ as a scale distortion. The realized Type I error follows at once,

$$\alpha_{\text{real}} = \Pr(|T| > t_{\nu, 1-\alpha/2}) \approx \Pr(|Z^* + \delta| > \kappa t_{\nu, 1-\alpha/2}), \quad (3)$$

and exactly four quantities appear in it. They are the four channels: the standardized bias δ , which decenters the statistic; the scale κ , which multiplies the rejection threshold; the degrees of freedom ν , which fix the critical value $t_{\nu, 1-\alpha/2}$; and the distribution of Z^* , the shape of the reference. When all four take their nominal values, $\delta = 0$, $\kappa = 1$, $\nu = \infty$, and $Z^* \sim N(0, 1)$, the expression collapses to $\Pr(|N(0, 1)| > z_{1-\alpha/2}) = \alpha$ and the test is exact; a departure in any one channel miscalibrates it. A second identity is worth recording, since the diagnosis turns on it: because $T \approx (Z^* + \delta)/\kappa$, the dispersion of the statistic satisfies

$$\text{SD}(T) \approx \text{SD}(Z^*) / \kappa, \quad (4)$$

so a statistic standard deviation below one is read at once as a scale inflation ($\kappa > 1$), a sub-normal reference shape, or both. One qualification attends the decomposition: equation (2) treats κ as a fixed mean ratio and neglects the sampling variability of $\widehat{\text{SE}}$ itself. That variability is not lost but carried by the degrees-of-freedom and reference-shape channels, with which the scale channel is therefore only approximately separable. The approximation is excellent when the model assigns many degrees of freedom to the standard error, as in the worked example below, and we return to its limits where the cluster-robust standard error, which carries far fewer, is taken up.

We measure the four channels by simulation as follows.

1. **Point-estimate bias** (δ). If $E(\hat{\theta}) \neq 0$ under H_0 the statistic is decentred, acquiring the non-centrality of a noncentral t ¹⁴. We measure μ , and hence $\delta = \mu/\sigma$, by the mean of $\hat{\theta}$ across null replicates.
2. **Standard-error calibration** (κ). If the model-based $\widehat{\text{SE}}$ does not estimate the true sampling standard deviation of $\hat{\theta}$, the statistic is rescaled. We measure κ as the mean model-reported standard error over the empirical standard deviation of the estimate across replicates. That unequal error variances or correlation among the errors distort the realized level of a test through this channel is a classical result of Box^{15,16}. A ratio above one indicates an over-stated standard error and a conservative test; a ratio below one the reverse.

3. **Degrees of freedom** (ν). If ν is too small the t reference has tails too heavy and the test is conservative; if too large, the reverse. The approximate degrees of freedom such a reference carries trace to Satterthwaite¹⁷ and Welch¹⁸. We measure this by recomputing the p-values against a normal reference and comparing the realized Type I error.
4. **Reference-distribution shape** (the law of Z^*). Even with $\delta = 0$, $\kappa = 1$, and ν correct, Z^* may not be standard normal. The consequences of such non-normality for the true level of a test were studied classically by Pearson¹⁹, Box²⁰, and Scheffé²¹, ch. 10. We measure this by the standard deviation of T under H_0 , which should be approximately one, and by the empirical distribution function of the p-values, which should be Uniform(0, 1).

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The individual channels are not new, and each carries a classical provenance. The decentering of the statistic by a non-null mean is the non-centrality of the non-central t distribution, whose properties were charted by Johnson and Welch¹⁴; as performance measures, the bias of the estimate and the ratio of the model-based to the empirical standard error are among the standard summaries of a simulation study codified by Morris, White, and Crowther²², who recommend that the model-based and empirical standard errors be reported and compared as a matter of course. The scale channel has the longest pedigree of the four: that unequal error variances and correlation among the errors push the realized significance level of a test away from its nominal value, precisely the mechanism by which a misspecified working covariance acts here, was established by Box in his two-part study of departures from the assumptions of the analysis of variance^{15,16}, the asymptotic counterpart being the inconsistency of the model-based standard error under the conditions of Eicker, Huber, and White³⁻⁵. The degrees-of-freedom channel rests on the approximate-distribution arguments of Satterthwaite¹⁷ and Welch¹⁸, refined for the mixed model by Kenward and Roger²³. The reference-shape channel, the departure of the law of Z^* from normality, is the subject of the classical robustness literature of Pearson¹⁹, Box²⁰, and Scheffé²¹, ch. 10, its formal small-sample anatomy given by the Edgeworth expansion of the studentised statistic²⁴, with the uniformity of the null p-value distribution following from the probability integral transform; the anatomy of the Wald statistic that organizes the four is itself standard¹³. What the decomposition contributes is not a new measure, nor the observation that any single assumption matters, but the demonstration that these four quantities, and exactly these four, compose into the single expression for α_{real} derived above. Two consequences follow. First, the channels are exhaustive: a Wald test is a centered statistic, a scale, a reference shape, and a degrees-of-freedom choice, and the realized error rate is a function of those and of nothing else, so a discrepancy absent from all four cannot exist. Second, because the channels enter the expression in distinct and separately measurable roles, they interact within α_{real} but are estimated independently, so measuring all four does not merely detect a miscalibration but localizes it, identifying which term

carries the fault. Since the channels have different causes and different remedies,
a diagnosis is in this sense a decomposition and not a checklist.

2.2 The working-covariance model ladder

The four-channel decomposition tells us how a test is miscalibrated. It does not tell us why. When the decomposition points to the standard-error channel, it is worth being precise about what the scale κ measures. Let the interaction estimate be written $\hat{\theta} = \ell' \hat{\beta}$, with ℓ the contrast vector that selects the interaction coefficient from the fixed-effects vector. A fixed-effect estimator that solves a generalized least squares system under a working covariance V_0 , when the data are in truth generated under a covariance V , has a model-based variance

$$v_0 = \ell'(X'V_0^{-1}X)^{-1}\ell, \quad (5)$$

while its true sampling variance is the sandwich form

$$v = \ell'(X'V_0^{-1}X)^{-1}(X'V_0^{-1}VV_0^{-1}X)(X'V_0^{-1}X)^{-1}\ell, \quad (6)$$

where X is the fixed-effects design and, the clusters being independent, every cross-product is a sum over clusters, $X'V_0^{-1}X = \sum_i X_i'V_{0i}^{-1}X_i$ and likewise $X'V_0^{-1}VV_0^{-1}X = \sum_i X_i'V_{0i}^{-1}V_iV_{0i}^{-1}X_i$ ^{5,6,25}. For the mixed model fitted by maximum or restricted maximum likelihood^{26,27}, the inconsistency of the model-based variance v_0 for the true variance v under a wrong V_0 is the misspecified-likelihood result of White²⁸, of which the sandwich form is the consistent counterpart. The scale of Section 2.1 is the ratio of these two quadratic forms, $\kappa^2 = v_0/v$. They coincide when $V_0 = V$, and for a generic contrast ℓ they do not otherwise; the scale channel is therefore not a finite-sample nuisance but a direct consequence of the working covariance: κ departs from one whenever $V_0 \neq V$, and no quantity of data removes the discrepancy.

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This is also what makes a model ladder informative. Each rung is a different working covariance V_0 fitted to the same data, so tracking κ along a graded sequence of working covariances, from the simplest to the production model, isolates the rung, that is, the covariance component, at which κ departs from one. Because every rung is fitted to the identical simulated data, the comparison is free of Monte Carlo noise between models.

2.3 The sample-size gradient

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A standard-error miscalibration may be one of two qualitatively different things. It may be a finite-sample artifact, in which case it attenuates as the sample grows and the working-covariance parameters are estimated more sharply; the small-sample

corrections of Kenward and Roger^{23,29} address miscalibration of this kind. Or it may be asymptotic, a consequence of a working covariance that does not match the data-generating covariance no matter how much data is collected, in which case it persists. The two are distinguished by running the decomposition across a range of cluster counts. Here a point of construction guides the reading. The scale κ of Section 2.2 is a functional of the per-cluster quantities (X_i, V_i, V_{0i}) alone, so it does not depend on the number of clusters and is, to leading order, constant along the gradient by construction; what a varying cluster count changes is the precision with which the variance parameters of V_0 are estimated, the very quantity whose finite-sample uncertainty the Kenward-Roger correction addresses, following Kackar and Harville³⁰. A gradient that is flat in the cluster count therefore does two things at once: it confirms that no spurious trend enters through parameter estimation, and it shows that the bias is not of the kind a finite-sample correction would remove.

Attenuation toward nominal calibration marks the finite-sample case; a flat profile marks the asymptotic case. The distinction matters, because a finite-sample artifact at a realistic sample size may be tolerable, whereas an asymptotic bias is a property of the method and will not be outgrown. The gradient varies the number of clusters and not the number of timepoints within a cluster; the latter is the axis along which the per-cluster covariance mismatch could itself change, and we return to it among the limitations.

3 Worked example: an interaction test under carryover

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The example we shall work through is the test of a biomarker-treatment interaction in the carryover-sensitivity simulation study of the pmsimstats-ng project³¹. Aggregated N-of-1 trials repeatedly cross each participant between an active treatment and a control condition, and the scientific question is whether a baseline biomarker moderates the treatment effect. Carryover, the persistence of drug effect into an off-drug period, is a long-standing concern in the analysis of crossover designs^{32,33}, and the study compares analysis-model strategies for accommodating it²². The reporting follows the ADEMP structure of Morris, White, and Crowther²².

3.1 The analysis model

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For a symptom score $S_{x,it}$ on participant i at timepoint t , the analysis model is

303 the linear mixed model

$$304 \quad S_{x,it} = \beta_0 + \beta_1 \text{bm}_i + \beta_2 t + \beta_3 X_{it} + \theta (\text{bm}_i \times X_{it}) + b_i + \varepsilon_{it}, \quad (7)$$

305 fitted by `nlme::lme` with a participant random intercept $b_i \sim N(0, \sigma_b^2)$ and a
306 continuous-time first-order autoregressive residual correlation, `corCAR1(form = ~`
307 `t | ptID)`. Here bm_i is the standardized biomarker, X_{it} a drug-exposure predictor,
308 and θ the interaction coefficient of interest. The continuous-time autoregressive
309 residual was adopted in the project for its clinical realism, nearby measurements
310 being more strongly correlated than distant ones, and for the numerical range it
311 affords¹¹. Whether that choice is also well calibrated for the fixed-effect interaction
312 test is precisely the question we shall address.

313 3.2 The null configuration

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316 Calibration is a null-hypothesis property, so the diagnosis is run where the
317 biomarker exerts no moderation, $c_{bm} = 0$, and the true interaction is therefore
318 $\theta = 0$. The primary cell is the reference configuration of the parent study: the
319 Hybrid N-of-1 design, $N = 70$ participants, an exponential carryover form with a
320 one-week half-life, and an autoregressive parameter of 0.7. A preliminary check es-
321 tablished that at $c_{bm} = 0$ the study's two data-generating architectures³⁴ collapse
322 to a bit-identical process, so the diagnosis need not stratify by architecture.

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325 The feature of the example that merits attention for the argument is the rela-
326 tionship between the working covariance and the truth. The simulated symptom
327 score is a sum of three latent factors, each following its own autoregressive process,
328 with timepoint-dependent variances, plus a participant-level baseline term. The
329 working covariance of the analysis model, one random intercept and one single-
330 parameter `corCAR1` residual, cannot reproduce that structure exactly. The exam-
331 ple is therefore an instance of the general situation the protocol is designed for:
332 a parametric working covariance that is plausible, conventional, and not exactly
333 correct.

334 4 Results

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337 The diagnosis was carried out at 5000 null replicates for the four-channel decom-
338 position and the model ladder, and at 1500 replicates per point for the sample-size
339 gradient. The cluster-robust remedy was verified at 3000 replicates. A rejection

Table 1: Four-channel decomposition at the primary null cell (Hybrid design, $N = 70$, $c_{bm} = 0$, 5000 replicates), for the three analysis specifications E1, E2, E3. The interaction estimate is effectively unbiased; the SE ratio exceeds one; the statistic is correspondingly compressed; the t and z references coincide because the degrees of freedom are large.

Spec	Mean estimate	SE ratio κ	SD(T)	Type I (t)	Type I (z)	Mean DF
E1	+0.032	1.069	0.932	0.033	0.034	487
E2	+0.039	1.097	0.909	0.029	0.029	487
E3	+0.033	1.061	0.939	0.035	0.036	486

rate estimated from n_{sim} replicates carries an MCSE of $\sqrt{p(1-p)/n_{\text{sim}}}$ ²²; at a Type I error near 0.03 this is approximately 0.0024 at 5000 replicates, 0.0031 at 3000, and 0.0044 at 1500, and the comparisons to the nominal level reported below are read against those tolerances. All replicate-level outputs are retained for inspection.

4.1 The conservatism is standard-error over-estimation

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Table 1 reports the decomposition for the three analysis specifications carried by the parent study, which differ only in the drug-exposure predictor X_{it} and are treated here as three instances of the same interaction test. The reading is consistent across all three. The interaction estimate is effectively unbiased: the null mean falls between +0.03 and +0.04 across the three specifications, on a coefficient whose sampling standard deviation is near unity, a displacement of about 0.035 of a standard deviation. A displacement of that size is statistically detectable at 5000 replicates but is negligible in magnitude, and in any case it points toward rejection and so cannot produce a conservative test. The degrees-of-freedom channel is excluded directly: the model assigns approximately 487 degrees of freedom to the interaction term, and recomputing the p-values against a normal reference moves the realized Type I error by about 0.001, as the near-identical t - and z -reference columns show.

The discrepancy is in the standard-error channel. The model-based standard error exceeds the empirical sampling standard deviation of the estimate by 6 to 10 percent, and the standard deviation of the test statistic agrees with the reciprocal of that ratio to within half a percent, as the identity $\text{SD}(T) \approx \text{SD}(Z^*)/\kappa$ of equation (4) with $\text{SD}(Z^*) = 1$ leads us to expect. With the other three channels inert, $\delta \approx 0$, ν large, and Z^* standard normal, the realized Type I error of equation (3) reduces to the closed form

$$\alpha_{\text{real}} \approx 2 \Phi(-\kappa t_{\nu, 1-\alpha/2}), \quad (8)$$

the probability that a standard normal exceeds the threshold $\kappa t_{\nu, 1-\alpha/2}$ rather than $z_{1-\alpha/2}$.

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371 Differentiating this closed form exposes the leverage of the scale channel. Near
 372 $\kappa = 1$, with ν large enough that $t_{\nu, 1-\alpha/2} \approx z_{1-\alpha/2} = 1.96$,

$$373 \quad \frac{d \alpha_{\text{real}}}{d \kappa} = -2 t_{\nu, 1-\alpha/2} \phi(\kappa t_{\nu, 1-\alpha/2}) \approx -0.23, \quad (9)$$

374 so to first order $\alpha_{\text{real}} \approx 0.05 - 0.23 (\kappa - 1)$. The leverage is best read as an elasticity:
 375 at $\kappa = 1$, $d \log \alpha_{\text{real}} / d \log \kappa \approx -4.6$, so each one percent of over-statement in the
 376 standard error removes about four to five percent of the nominal Type I error. A
 377 six-to-ten percent over-statement therefore depresses the Type I error into the low
 378 0.03 range, and it is in this sense that a numerically modest miscalibration of the
 379 standard error constitutes a material miscalibration of the test. For the standard-
 380 error ratios of Table 1 this predicts a Type I error of 0.031 to 0.037 across the
 381 three specifications, matching the observed 0.029 to 0.033 (at $\alpha = 0.05$) to within
 382 Monte Carlo error. The scale channel alone, with no contribution from the other
 383 three, reproduces the conservatism.

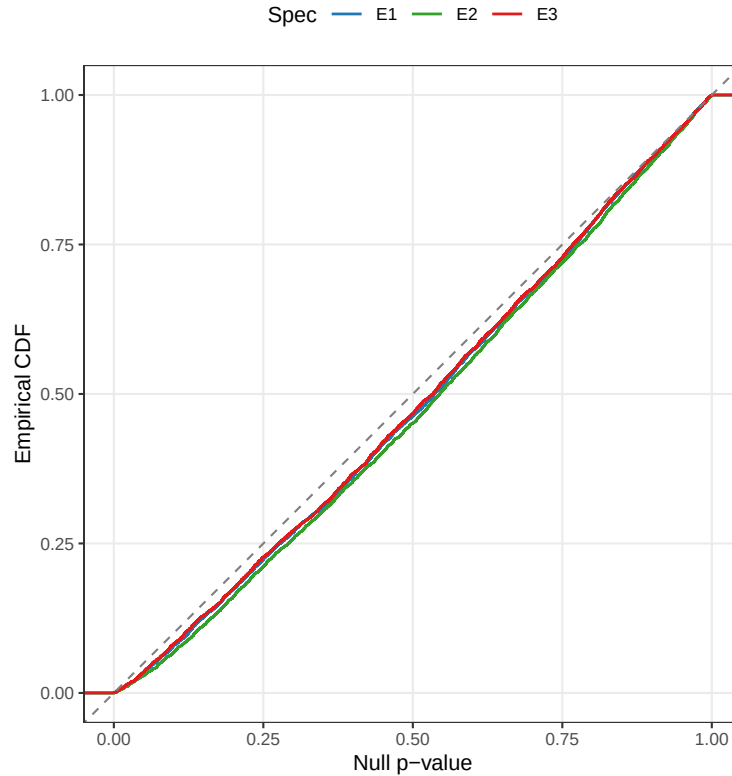


Figure 1: Empirical distribution function of the null p-values at the primary cell, against the Uniform(0,1) diagonal. All three specifications bow uniformly below the diagonal, the signature of a global scale compression rather than a tail anomaly.

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Table 2: Working-covariance model ladder at the primary null cell (5000 replicates), for specifications E1 and E2. The same datasets are refitted under each structure. M1, a random intercept alone, is well calibrated; M3, the production model, adds a corCAR1 residual and is conservative.

Model	Structure	κ (E1)	κ (E2)	Type I (E1)	Type I (E2)
M0	OLS, no within-cluster correlation	1.323	1.289	0.010	0.012
M1	Random intercept only	0.989	0.972	0.049	0.055
M2	corCAR1 residual only	1.023	1.070	0.043	0.036
M3	Random intercept + corCAR1 (production)	1.068	1.097	0.033	0.029

Figure 1 confirms the nature of the miscalibration. The empirical distribution function of the null p-values lies below the diagonal across the whole unit interval and for all three specifications. A test that failed only in its tails would depart from the diagonal only near zero; the uniform bowing seen here is the signature of a global scale compression, which is precisely what an inflated standard error produces. The positive-definiteness correction available to the data generator was not triggered in any of the cell’s paths and is excluded as a contributor.

4.2 The corCAR1 residual layer is the cause

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The decomposition tells us that the standard error is over-estimated. The model ladder, Table 2, tells us which part of the model does the over-estimating. Four working covariances were fitted to the identical null datasets: ordinary least squares with no within-cluster term (M0), a random intercept alone (M1), a corCAR1 residual alone (M2), and the production combination of a random intercept with a corCAR1 residual (M3).

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The result is unambiguous. M1, the random intercept with no residual correlation, is well calibrated: its standard-error ratio sits between 0.97 and 0.99 and its Type I error between 0.049 and 0.055. M3, which adds the corCAR1 residual on top of that random intercept, is conservative: the ratio rises to between 1.07 and 1.10 and the Type I error falls to between 0.029 and 0.033. The corCAR1 residual-correlation layer is therefore the proximate cause of the conservatism. It is worth being precise about what this does and does not say. It does not say that the corCAR1 residual is wrong in any absolute sense; the simulated factors are genuinely autocorrelated, so a residual correlation term is capturing a real feature of the data. What it says is that adding that term to a model which already carries a random intercept degrades the calibration of the fixed-effect interaction test.

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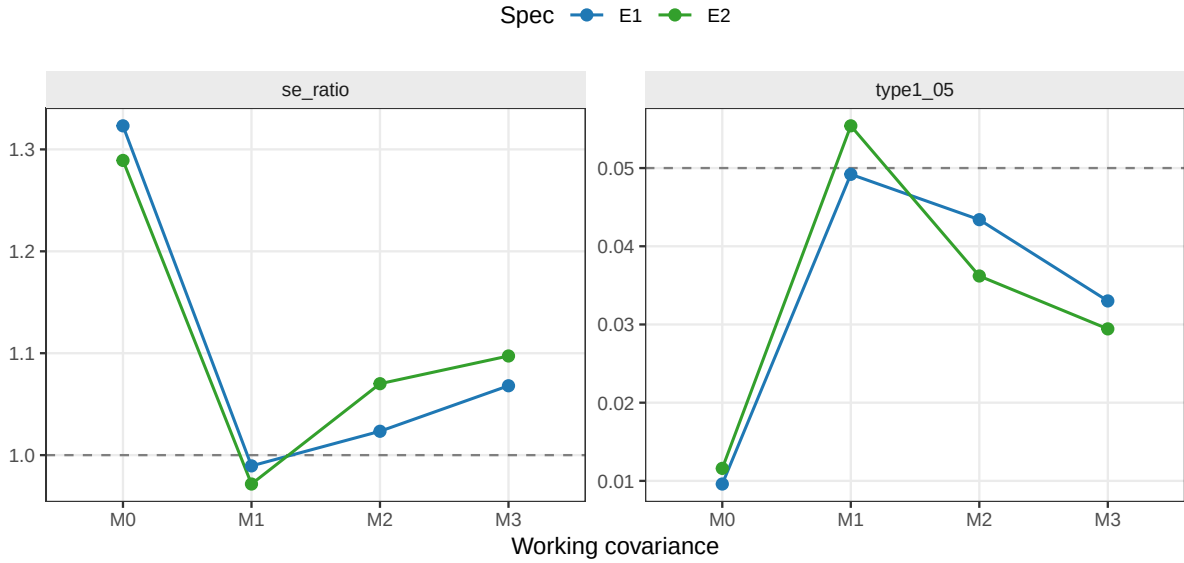


Figure 2: Interaction standard-error calibration across the model ladder. Left, the ratio of model-reported to empirical standard error; right, the realized Type I error. The dashed lines mark perfect calibration. The random intercept alone (M1) sits on both reference lines; the production model (M3) does not.

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417 Two further features of the ladder, visible in Figure 2, are worth recording. The
 418 first is that M0, ordinary least squares, is not anti-conservative, as the familiar
 419 warning that ignoring positive within-cluster correlation under-states standard er-
 420 rors would suggest. It is strongly conservative, with a standard-error ratio near
 421 1.3. The reason is instructive: the interaction here is a within-participant contrast,
 422 and ordinary least squares charges it the full residual variance, between-participant
 423 and within-participant combined, when the error relevant to a within-participant
 424 contrast is only the within-participant part. In the idealised case of a pure within-
 425 participant contrast with exchangeable errors the inflation is the variance ratio
 426 $\kappa_{\text{OLS}}^2 \approx (\sigma_b^2 + \sigma_\epsilon^2)/\sigma_\epsilon^2$, the total residual variance over its within-participant com-
 427 ponent, exceeding one by the random-intercept variance σ_b^2 . The familiar warning,
 428 the Moulton pitfall, concerns between-cluster effects³⁵; for a within-cluster effect
 429 the sign of the OLS error is reversed. The distinction between within- and between-
 430 cluster covariate effects that governs this sign is the one drawn by Neuhaus and
 431 Kalbfleisch³⁶, and the inflation of the ordinary-least-squares variance under intra-
 432 cluster correlation is the two-stage-sampling result of Scott and Holt³⁷. The second
 433 feature is that the corCAR1 residual without a random intercept (M2) is intermedi-
 434 ate. Taken together, the ladder shows that calibration is recovered by the random
 435 intercept alone and removed by the corCAR1 addition, and that the magnitude
 436 and even the sign of the miscalibration is a property of the working covariance,
 437 not a universal constant.

Table 3: Sample-size gradient under the production model M3 (1500 replicates per point). The standard-error ratio κ and the Type I error (T1) are shown for each specification. Neither trends toward its nominal value as N increases.

N	κ E1	κ E2	κ E3	T1 E1	T1 E2	T1 E3
35	1.072	1.103	1.064	0.038	0.028	0.039
70	1.056	1.083	1.050	0.043	0.031	0.045
140	1.086	1.098	1.079	0.041	0.033	0.042
280	1.052	1.070	1.046	0.037	0.028	0.035

4.3 The conservatism is asymptotic, not a small-sample artifact

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If the conservatism were a finite-sample artifact, it would diminish as the sample grew and the working-covariance parameters were estimated more precisely. Table 3 and Figure 3 show that it does not. Across an eightfold range of sample size, from $N = 35$ to $N = 280$ participants, the standard-error ratio holds between 1.05 and 1.10 with no trend toward one, and the Type I error holds between 0.028 and 0.045 with no trend toward 0.05. At the largest sample size the production specification still rejects the true null at 0.028. This is what the construction of Section 2.3 predicts: the scale κ is a per-cluster functional and cannot trend with the cluster count, so the flat profile’s work is to confirm that no finite-sample trend of the Kenward-Roger kind is present either. The conservatism is asymptotic. It appears to be a property of the M3 working covariance and will not be outgrown.

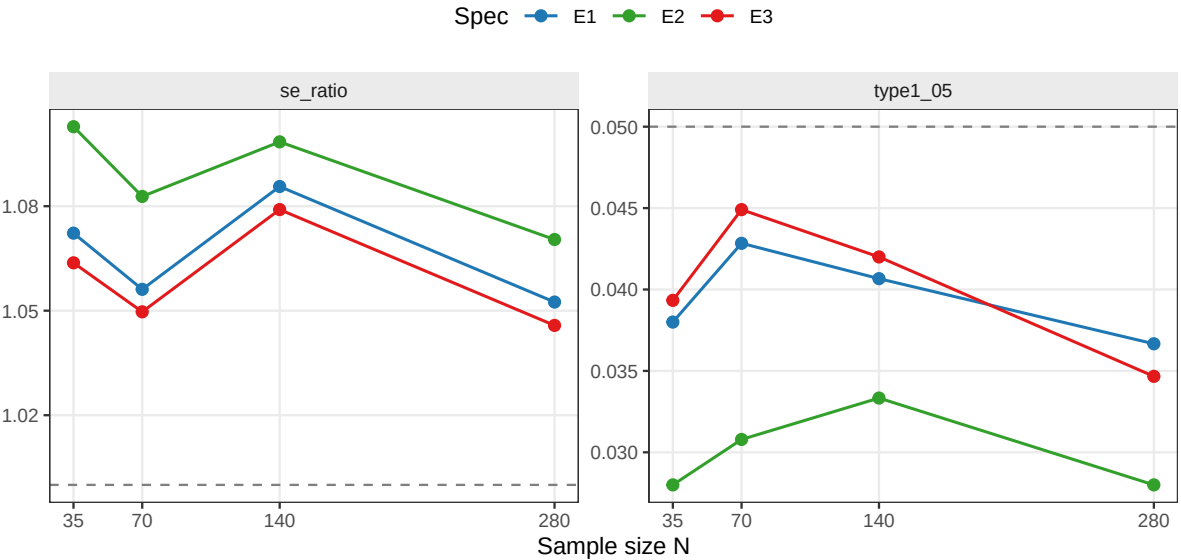


Figure 3: Type I error and standard-error ratio against sample size under the production model M3. Neither panel trends toward its dashed reference line as the number of participants increases from 35 to 280.

Table 4: Cluster-robust remedy at the primary null cell (3000 replicates). The model-based standard error of the production model is replaced by a CR2 bias-reduced cluster-robust standard error clustered on participant, with Satterthwaite degrees of freedom. The CR2 standard-error ratio is approximately one and the Type I error approximately 0.05.

Spec	κ model-based	κ CR2	Type I model	Type I CR2	CR2 df
E1	1.062	0.978	0.036	0.048	23.7
E2	1.096	1.000	0.029	0.045	23.7
E3	1.053	0.976	0.039	0.049	23.7

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The explanation that is consistent with all three results is a working-covariance mismatch rather than a small-sample effect. The simulated symptom score is a mixture of autoregressive factors with unequal, timepoint-dependent variances, plus a participant baseline term. The M3 working covariance, a single random intercept and a single-parameter corCAR1 residual, cannot match a mixture of autoregressive processes exactly. Increasing the number of participants estimates the parameters of that working covariance ever more sharply, but it does not change the fact that the covariance being estimated is, in a sense, the wrong one, and so the model-based standard error converges to a biased target. The bias persists because it is a bias of the model, not of the estimate of the model's parameters.

4.4 A cluster-robust standard error restores calibration

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The diagnosis points to a remedy that does not require abandoning the corCAR1 model. If the model-based standard error is biased because the working covariance is wrong, the natural replacement is a standard error that is consistent whatever the working covariance happens to be^{5,6,38}. We replace the model-based standard error of the interaction with a cluster-robust standard error, clustered on participant, in the bias-reduced CR2 form of Bell and McCaffrey⁸, with the Satterthwaite degrees of freedom of Pustejovsky and Tipton^{10,17}. The CR2 adjustment and its associated small-sample test were designed for precisely the regime we face here, a moderate number of clusters, and are implemented for lme objects in the clubSandwich package^{39,40}.

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It is worth setting down the construction. Let $\tilde{e}_i$ collect the residuals of cluster i from the fitted model and let H_{ii} be the corresponding block of the hat matrix, both taken in the metric defined below. A cluster-robust variance replaces

the model-based variance of $\hat{\theta}$ by a quadratic form in the empirical cluster cross-products,

$$\hat{V}_{\text{CR}} = (\tilde{X}'\tilde{X})^{-1} \left(\sum_i \tilde{X}'_i A_i \tilde{e}_i \tilde{e}'_i A_i \tilde{X}_i \right) (\tilde{X}'\tilde{X})^{-1}, \quad (10)$$

where $\tilde{X}$ and $\tilde{e}$ denote the design and residuals in the metric of the working covariance, that is, after the generalized-least-squares transformation $\hat{V}_0^{-1/2}$; for ordinary least squares the transformation is the identity and the familiar form is recovered, and the family CR0, CR1, CR2, CR3 is indexed by the adjustment matrices A_i ^{9,10}. The unadjusted choice $A_i = I$ is consistent as the number of clusters grows but biased downward at moderate cluster counts, and its own sampling variability is substantial in that regime^{41,42}; the jackknife choice $A_i = (I - H_{ii})^{-1}$ over-corrects and is, in the marginal setting, the estimator of Mancl and DeRouen⁷. The CR2 choice sets $A_i = (I - H_{ii})^{-1/2}$, the symmetric square root that renders $\hat{V}_{\text{CR}}$ exactly unbiased for the true variance when the working covariance is in fact correct; it is the cluster generalization of the HC2 estimator of MacKinnon and White⁴³. The working-covariance metric for mixed models follows Pustejovsky and Tipton¹⁰, and the pairing with a matching small-sample reference is recommended by Imbens and Kolesar⁴⁴. Because $\hat{V}_{\text{CR}}$ is itself a quadratic form in approximately normal residuals, its null distribution is approximated by a scaled χ^2 , and the Satterthwaite degrees of freedom are recovered by matching its first two moments, $\nu = (\sum_g \lambda_g)^2 / \sum_g \lambda_g^2$ in the eigenvalues λ_g of the matrix defining that form^{10,17}.

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Table 4 reports the verification. The model-based standard-error ratio reproduces the conservatism, between 1.05 and 1.10. The CR2 cluster-robust ratio is close to one, between 0.97 and 1.00, and the realized Type I error rises from approximately 0.03 under the model-based test to between 0.045 and 0.049 under the CR2 test, within Monte Carlo error of the nominal 0.05. The Satterthwaite degrees of freedom for the cluster-robust test are approximately 24, far below the model-based 487 and well below the participant count of 70, since they measure the heterogeneity of cluster leverage on the interaction contrast rather than a count of clusters, the effective number of clusters in the sense of Carter, Schnepel, and Steigerwald^{44,45}; the small-sample reference is doing real work, and a naive normal reference for the sandwich statistic would itself be mildly anti-conservative^{9,46}.

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Figure 4 shows the same result as a distribution function. The model-based p-values bow below the diagonal, as in Figure 1; the cluster-robust p-values lie on it. The point estimate and the corCAR1 model are unchanged; only the standard error of the interaction, and the reference against which it is tested, are replaced.

4.5 The remedy at an alternative cell

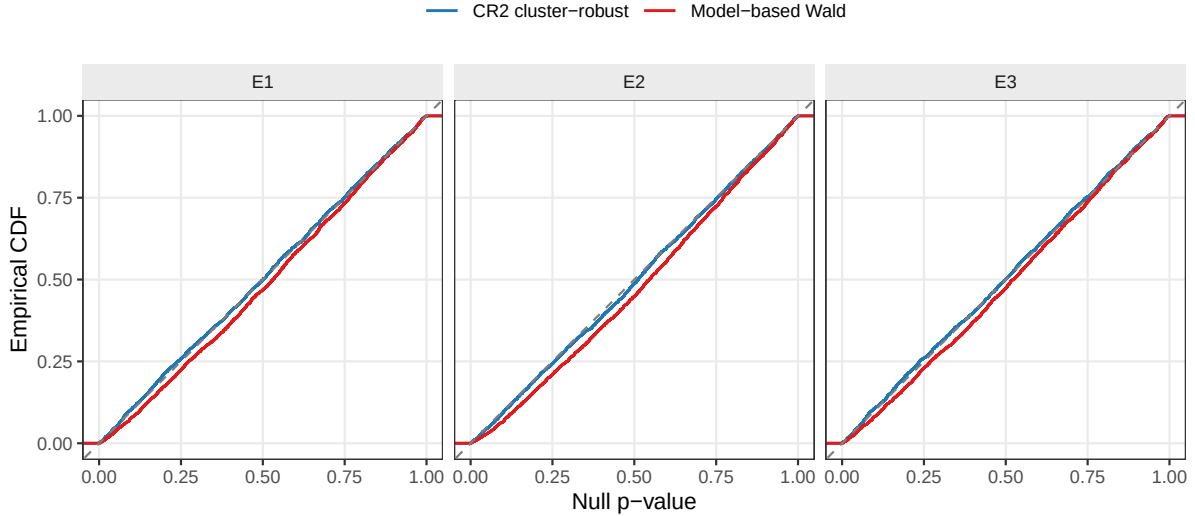


Figure 4: Null p-value distribution functions under the model-based and CR2 cluster-robust tests, by specification. The model-based curves bow below the diagonal; the CR2 curves lie on it.

Table 5: The cluster-robust remedy at an alternative cell (the covariance-moderation architecture, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, 3000 replicates). Power is the rejection rate of the interaction test; the standard-error ratio κ is as defined for the null. The model-based test is conservative here too, and the CR2 test recalibrates it.

Spec	κ model-based	κ CR2	Power model	Power CR2
E1	1.124	0.986	0.757	0.795
E2	1.152	0.988	0.843	0.878
E3	1.119	0.989	0.755	0.788

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525 The diagnosis to this point has been conducted under the null, where calibration
526 is defined. A reader of a comparative study will, however, press a further question.
527 The purpose of the parent study is to rank analysis specifications by power, and if
528 a recalibration of the standard error raises the power of every specification, does it
529 also disturb their order? We checked this directly at one alternative cell, the parent
530 study's reference configuration with a non-null biomarker moderation, $c_{bm} = 0.45$,
531 retaining the Architecture-B Hybrid design at $N = 70$.

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534 Three things follow from Table 5. First, the standard-error over-estimation is not
535 confined to the null: at the alternative cell the model-based ratio is 1.12 to 1.15,
536 if anything slightly larger than the 1.06 to 1.10 seen under the null. Second, the
537 CR2 cluster-robust standard error recalibrates the alternative cell as it did the null,

bringing the ratio to 0.99 for all three specifications. Third, and most consequential for a comparative study, the recalibration is close to a uniform level shift. It raises the power of every specification by roughly three to four percentage points, and the gaps between specifications are left intact: the advantage of the leading specification E2 over E1 is +0.086 under the model-based test and +0.082 under the cluster-robust test, and over E3 it is +0.088 and +0.089. The ranking, and the magnitude of the leading specification's advantage, survive the recalibration unchanged.

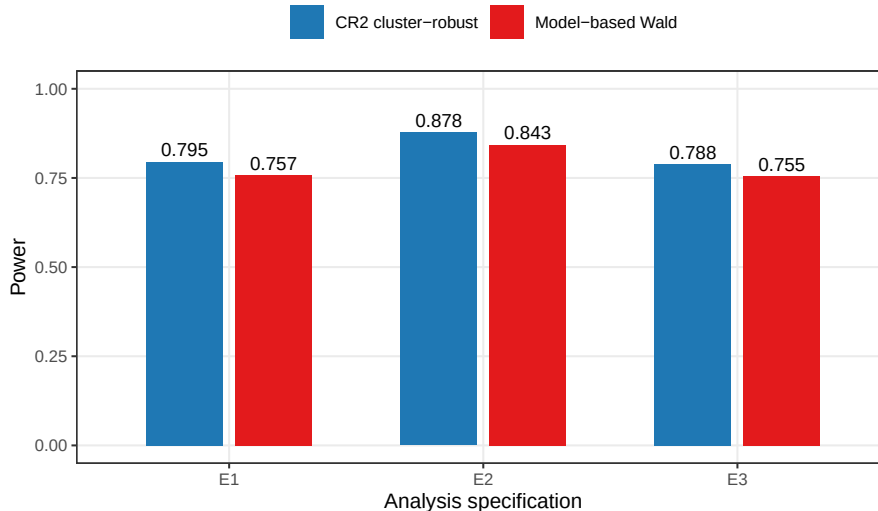


Figure 5: Power of the three analysis specifications at the alternative cell, under the model-based and CR2 cluster-robust tests. The recalibration raises every specification by a near-uniform increment and preserves their order.

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The lesson generalizes beyond the example. A cluster-robust standard error corrects the absolute error rate, the Type I error under the null and the power under the alternative alike, but because the correction acts on the standard error and not on the point estimate, and because the standard-error miscalibration here is relatively similar across the specifications being compared, it shifts the level without reordering the contrast. A comparative study whose claim is a ranking is therefore robust to the recalibration, even where its absolute power figures, as reported under the model-based test, are mildly pessimistic.

5 Discussion

5.1 The result is not about autoregressive correlation

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It would be a mistake to read this paper as a caution against the continuous-

time autoregressive residual in particular. The corCAR1 term is the component the model ladder happened to implicate in this example, but the mechanism is general. The model ladder is itself the evidence: it fitted four working covariances and obtained four different standard-error ratios, from 1.3 under ordinary least squares to 1.0 under a random intercept to near 1.1 under the production model. The miscalibration is a property of the match between the working covariance and the truth, and any parametric residual or random-effects structure, compound symmetry, an unstructured form, a random slope, a spatial correlation, is capable of producing it; the effect of a serial-correlation structure on the variance of a least-squares contrast was examined classically by Watson⁴⁷, and the robustness of mixed-model fixed-effect inference to a misspecified covariance by Verbeke and Lesaffre⁴⁸. That a misspecified working covariance leaves the model-based fixed-effect standard error inconsistent is, moreover, not a new claim; it is the Eicker-Huber-White result carried into the mixed-model setting, and for a likelihood fit the misspecified-maximum-likelihood result of White^{3-5,28,49}. What the worked example contributes is a quantified, decomposed instance in a realistic longitudinal interaction test, with two features that are not obvious in advance.

5.2 Two features that are not obvious in advance

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The first is the direction. The sandwich literature is usually motivated by anti-conservative tests: ignored positive correlation under-states standard errors, and the robust estimator corrects an inflated rejection rate downward. Here the model-based test is conservative, and the cluster-robust estimator corrects the rejection rate upward. The ladder explains why the intuition can mislead. The effect under test is a within-cluster contrast, and for a within-cluster contrast the relationship between a misspecified covariance and the standard-error bias need not have the sign that the between-cluster intuition supplies. An analyst who reasons that ‘my standard errors are conservative because I have modeled the correlation’ has the logic backward in this case: modeling the correlation, through the corCAR1 term, is what introduced the conservatism, and the less elaborate model was the calibrated one.

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The second feature is that the bias is asymptotic. It would have been straightforward to attribute the conservatism to moderate sample size, and an earlier informal reading of the parent study did so. The sample-size gradient refutes that reading directly. This matters for how the problem is classified. The small-sample corrections of Kenward and Roger^{23,29} adjust the fixed-effect standard error and its degrees of freedom for the finite-sample uncertainty in the estimated variance parameters, operationalizing the variance-underestimation result of Kackar and

Table 6: Kenward-Roger versus Satterthwaite degrees of freedom at the primary null cell (2000 replicates), applied through the `mrmr` package to two marginal working covariances: a continuous-time autoregressive (spatial-exponential) covariance, the marginal counterpart of the `corCAR1` residual layer, and an unstructured covariance. Values are averaged over the three specifications. κ is the mean model standard error over the empirical standard deviation of the estimate. On the one-parameter autoregressive covariance the Kenward-Roger correction is inert; on the thirty-six-parameter unstructured covariance it corrects a genuine finite-sample anti-conservatism. Computed with the `mrmr` package (version 0.3.17).

Covariance	Method	κ	Mean df	Type I
AR(1), 1 parameter	Satterthwaite	1.075	476	0.036
AR(1), 1 parameter	Kenward-Roger	1.076	476	0.036
Unstructured, 36 parameters	Satterthwaite	0.892	164	0.076
Unstructured, 36 parameters	Kenward-Roger	0.955	164	0.059

Harville³⁰; they are the right instrument for a finite-sample artifact. They are not the right instrument here, because the bias is not a finite-sample artifact: it does not vanish as the variance parameters are estimated with unlimited precision. A Kenward-Roger correction applied to a misspecified working covariance corrects the finite-sample term and leaves the asymptotic mismatch untouched.

We confirmed this directly. Because the Kenward-Roger correction is not implemented for the production `corCAR1` mixed model in standard software, we applied it through the `mrmr` package⁵⁰ to the marginal counterpart of the implicated autoregressive layer, a continuous-time spatial-exponential working covariance, with an unstructured covariance fitted alongside for contrast (Table 6). The marginal autoregressive fit reproduces a conservatism of magnitude comparable to the production model's, with some variation across specifications, so the comparison across fitting engines is qualitative rather than exact. On the autoregressive covariance the correction was numerically inert: it altered the interaction standard error by about one tenth of one percent, holding the standard-error ratio at $\kappa \approx 1.08$ and the Type I error near 0.036, because a covariance carrying a single correlation parameter beyond its variance offers almost no finite-sample uncertainty for the correction to absorb. On the thirty-six-parameter unstructured covariance the same correction was far from inert, lifting the standard-error ratio from 0.89 to 0.96 and the Type I error from 0.076 toward 0.059. The magnitude of the Kenward-Roger adjustment thus tracks the number of covariance parameters, and for a parsimonious autoregressive working covariance it is too small to offset a six-to-ten percent bias. The cluster-robust standard error addresses the asymptotic mismatch directly, which is why it is the appropriate remedy.

5.3 When the problem should be expected

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Table 7: Marginal (GEE) check at the primary null cell, with an AR(1) working correlation (5000 replicates). The naive sandwich standard-error ratio κ and Type I error are shown beside the Mancl-DeRouen (MD) bias-corrected versions. The naive sandwich is mildly anti-conservative; the bias correction restores approximate calibration.

Spec	κ naive	κ MD	Type I naive	Type I MD
E1	0.955	0.992	0.068	0.055
E2	0.963	0.994	0.067	0.058
E3	0.955	0.991	0.069	0.055

629 The conditions for the problem are not exotic. They are a fixed-effect interaction
630 tested in a mixed model; a parametric working covariance that is plausible but
631 not known to be exactly correct, which is to say almost every parametric working
632 covariance in applied use; and an interaction that is partly or wholly a within-
633 cluster contrast. Wherever those conditions hold, the calibration of the model-
634 based Wald test is an open question, and the staged protocol of Section 2 answers
635 it at the cost of a simulation. We would suggest, as a matter of routine practice,
636 that an interaction test reported from a mixed model with a non-trivial working
637 covariance be accompanied either by a calibration check of this kind or by a cluster-
638 robust standard error, which removes the question by being agnostic to the working
639 covariance. This accords with a broader case for giving robust and marginal
640 methods more routine consideration alongside the conditional mixed model⁵¹.

641 5.4 The marginal-model alternative

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644 The remedy adopted above keeps the conditional mixed model and adds a cluster-
645 robust standard error after the fact. A reader may reasonably ask whether the
646 marginal route, generalized estimating equations, would not be more natural, since
647 the method carries the sandwich variance from the outset rather than bolting it
648 on^{6,52}. It would. For a linear model with an identity link the marginal and con-
649 ditional fixed-effect coefficients coincide⁵³, so a generalized estimating equations
650 analysis estimates the same interaction and answers the same question; the work-
651 ing correlation, exchangeable or autoregressive, governs only the efficiency of the
652 point estimate and not the validity of the inference.

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655 We checked the marginal route at the primary null cell, fitting the generalized
656 estimating equations counterpart of the analysis model with an AR(1) working
657 correlation; Table 7 reports the result. The naive sandwich standard error proved
658 mildly anti-conservative, the small-sample behavior the literature describes for a
659 moderate number of clusters^{42,54}: across the three specifications it under-estimated

the sampling standard deviation by roughly four percent, and the realized Type I error was 0.067 to 0.069. A bias-corrected sandwich of the kind introduced by Mancl and DeRouen⁷ removed most of the discrepancy, bringing the standard-error ratio to within one percent of one and the Type I error to 0.055 to 0.058. The marginal route is therefore viable, but it carries the same lesson as the conditional one: at the cluster counts typical of aggregated N-of-1 work the sandwich variance itself needs a small-sample correction⁴¹. The two corrected tests err in opposite and relatively modest directions, the conditional CR2 test slightly conservative and the marginal Mancl-DeRouen test slightly anti-conservative, and either is a defensible calibrated analysis. The choice between them is practical rather than principled: the conditional route preserves an existing mixed-model pipeline and changes only the standard error, while the marginal route is conceptually cleaner but a more thoroughgoing change of model.

5.5 Limitations

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Several limitations bound the scope of the evidence, and we shall endeavor to state them plainly. The diagnosis was carried out for one data-generating family, one estimand, and one production working covariance; the asymptotic conservatism is demonstrated for the corCAR1-plus-random-intercept model, and the generality beyond that rests on the model ladder and on established theory rather than on a sweep of every covariance structure. A broader empirical base, fitting an unstructured or compound-symmetry analysis to the same data, would strengthen the general claim, and is a natural extension. In the same spirit, the sample-size gradient varied the number of participants while holding the within-participant design fixed; the number of timepoints per cluster, the axis along which the per-cluster working-covariance mismatch could itself grow or shrink, was not varied, and a gradient in cluster size is a further natural extension. The cluster-robust remedy is not without cost: the Satterthwaite degrees of freedom for the CR2 test are approximately 24 against a model-based 487, so the robust test spends precision to buy validity, and at a genuinely small number of clusters that price would rise. The principal diagnosis was conducted at the null, where calibration is defined; the consequence for power was checked at a single alternative cell (Section 4.5), where the recalibration proved a near-uniform level shift that preserved the ranking of specifications. Whether that uniformity holds across the whole factorial, rather than at the one cell examined, was not established, and that question warrants further investigation. Downstream studies should therefore not extend the cluster-robust ranking-invariance or the conservative-lower-bound interpretation beyond the covariance-moderation architecture, Hybrid, $N = 70$ cells validated here, in particular not to crossover designs or to the mean-moderation architecture. Finally, the small positive point-estimate bias noted in Section 4.1, while negligible in magnitude and irrelevant to the conservatism, is a genuine finite-sample feature

702 of the estimator that a separate examination might characterize.

703 6 Conclusions

704 In conclusion, six points are to be emphasized. First, the model-based Wald test of a fixed-
705 effect interaction in a linear mixed model can be materially miscalibrated when the parametric
706 working covariance does not match the data-generating covariance. In the worked example
707 the miscalibration is a conservative one: the model-based standard error of the interaction is
708 over-estimated by 6 to 10 percent and the realized Type I error is approximately 0.03 against
709 a nominal 0.05.

710 Second, the four-channel decomposition localizes the discrepancy to the standard-error chan-
711 nel and excludes point-estimate bias, the degrees-of-freedom rule, and the positive-definiteness
712 correction.

713 Third, the working-covariance model ladder localizes the cause to the corCAR1 residual layer.
714 A random intercept alone is well calibrated; adding the corCAR1 residual removes the cali-
715 bration.

716 Fourth, the sample-size gradient shows the miscalibration is asymptotic. It does not attenuate
717 from $N = 35$ to $N = 280$, and so it is a property of the working covariance and not a small-
718 sample artifact amenable to a Kenward-Roger correction; applied empirically, the Kenward-
719 Roger adjustment proved numerically inert on the parsimonious autoregressive covariance
720 while correcting a genuine finite-sample anti-conservatism on an unstructured one.

721 Fifth, a CR2 bias-reduced cluster-robust standard error, clustered on participant and referred
722 to a Satterthwaite distribution, restores the standard-error ratio to approximately one and
723 the Type I error to approximately 0.05, without altering the point estimate or the analysis
724 model. At an alternative cell the same recalibration is a near-uniform level shift in power that
725 preserves the comparative ranking of specifications.

726 Sixth, and in our view most consequential for routine practice, the staged protocol, comprising
727 the four-channel decomposition, the model ladder, and the sample-size gradient, is a reusable
728 instrument for any mixed-model interaction test whose calibration is in doubt. We recommend
729 that interaction tests reported from mixed models with a non-trivial working covariance be
730 accompanied by a calibration check of this kind, or by a cluster-robust standard error that is
731 agnostic to the working covariance.

732 7 Reproducibility

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735 The diagnosis is driven by seven scripts under analysis/scripts/carryover-sensitivity/:
736 diagnose-null-type1.R for the four-channel decomposition, diagnose-null-type1-stages46.R
737 for the model ladder and the sample-size gradient, diagnose-null-type1-cr2.R
738 for the cluster-robust verification at the null, diagnose-alt-cell-cr2.R for

the alternative-cell check, `diagnose-null-type1-gee.R` for the GEE marginal-model check, and `diagnose-null-type1-kr.R` for the Kenward-Roger check, with `check-null-architecture-equivalence.R` establishing the architecture-invariance of the null. Each writes a replicate-level `.rds` file into the `output/` subdirectory, and this manuscript reads those files and regenerates every table and figure inline. The cluster-robust standard errors use the `clubSandwich` package, and the Kenward-Roger and Satterthwaite checks use the `mrmr` package⁵⁰. The examination plan that preceded the manuscript is retained alongside it as `type1-examination-plan.md`.

8 References

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